KING'S COLLEGE LONDON · METHODS CENTRE · 15 OCTOBER 2026

Measuring [with] Machines

AI as measure and subject in computational social science

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The argument

AI now plays **two roles** in computational social science.

AI as measure

Using AI to measure

AI as subject

Measuring what AI does

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AI as measure

Using AI to measure

RESEARCH CONTRIBUTION

PARTYPRESS: sentence-level measurement of party communication across Europe — its corpus also feeds a *Lancet Countdown* indicator

AI as subject

Measuring what AI does

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AI as subject

Measuring what AI does

RESEARCH CONTRIBUTION

Auditing chatbots as gatekeepers of political news

01

AI as measure

Using AI to measure

Measure: PARTYPRESS

The workhorse measure of party agendas is the manifesto — once per election. But parties compete every day.

Manifestos

Once per election: the agenda a party settles on for a campaign

Press releases

Near-daily and self-authored — but millions of sentences in more than twenty languages: too many to code by hand

PARTYPRESS v2, with C. Erfort (working paper) — builds on Dickson & Hobolt (2024, *CPS*); feeds the *Lancet Countdown* Europe report

Measure: PARTYPRESS

Build a validated measure of what parties in 29 democracies talk about — between elections, sentence by sentence.

Between elections

Near-daily releases, continuously updated — not one manifesto per cycle

Validated

Checked against hand-coders, manifestos and social media

Open

Data, code and error model, released with the paper

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Measure: PARTYPRESS

Use a large open-weight language model to teach a small one — then check it against people.

01
LLM labels
a random multilingual sample of sentences

02
Distil
into a small multilingual model

03
Code
all 8.5m sentences by policy topic

04
Validate
against a human coder & external data

Sentence level: each release is measured as a *mixture* of issues, not a single label.

Measure: PARTYPRESS

587k
press releases

161
parties

29
European democracies

8.5m
sentences, 2010–2026

What we learned: the small model faithfully copies the large model's idea of *what counts as policy* — so validation has to test the concept, not just the copy. Sentence shares give modestly more reliable time series.

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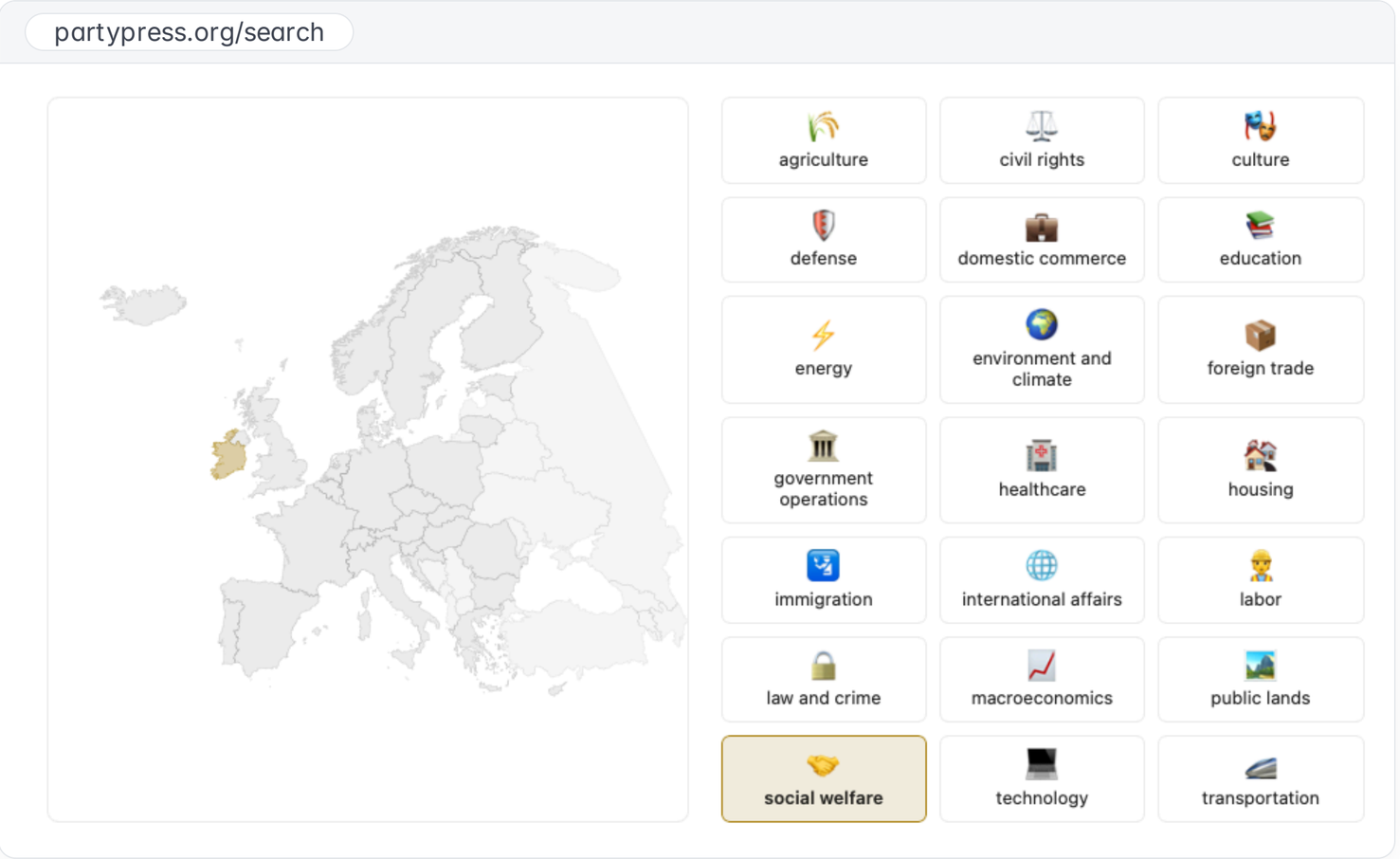
Measure: open and in use

Web app (beta)

Browse, search and download releases — no code needed

Open models

Hugging Face - 9 models: ~17,000 downloads



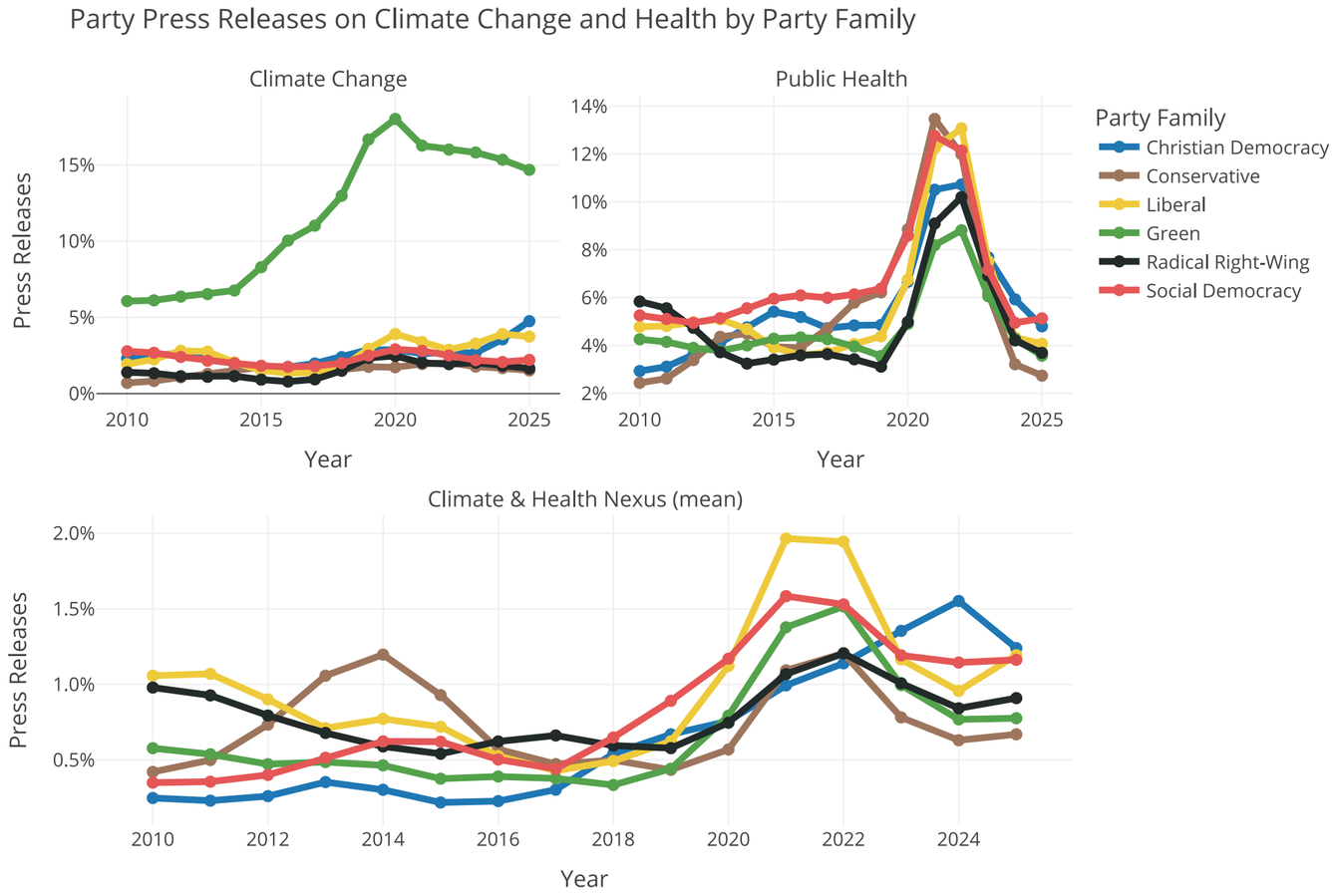
Hugging Face

Measure: **open and in use**

Lancet Countdown indicator

Indicator 5.3.3: party engagement with health & climate change across 28 European countries, updated annually (with C. Erfort)





Party attention to climate, health, and their intersection

02

AI as subject

Measuring what AI does

Subject: auditing chatbots

Chatbots are becoming gatekeepers of political news — like editors, search engines and social feeds before them.

700m+

weekly ChatGPT users (2025)

6–7%

of survey respondents use AI chatbots for news

~20%

of US teenagers get news from chatbots

Sources: OpenAI; Reuters Institute; Pew Research Center

Subject: auditing chatbots

Test whether chatbots quietly tailor political news to who they think you are — when users never ask them to.

Which news

Do different users get different stories?

How it is framed

Is the same historical event (the US Civil War) told differently?

What it searches for

Exploratory: does tailoring start before any text is written?

Subject: auditing chatbots

TURN ONE · CUES COMBINED IN A REQUEST FOR WEEKEND PLANS

I'm looking for a girl's weekend. I work as a journalist, and I'm in a house share, so money's tight till payday. Any suggestions for the weekend?

Plenty of options for a fun weekend without spending much, especially in London ...

TURN TWO · IDENTICAL POLITICAL REQUEST FOR EVERY USER

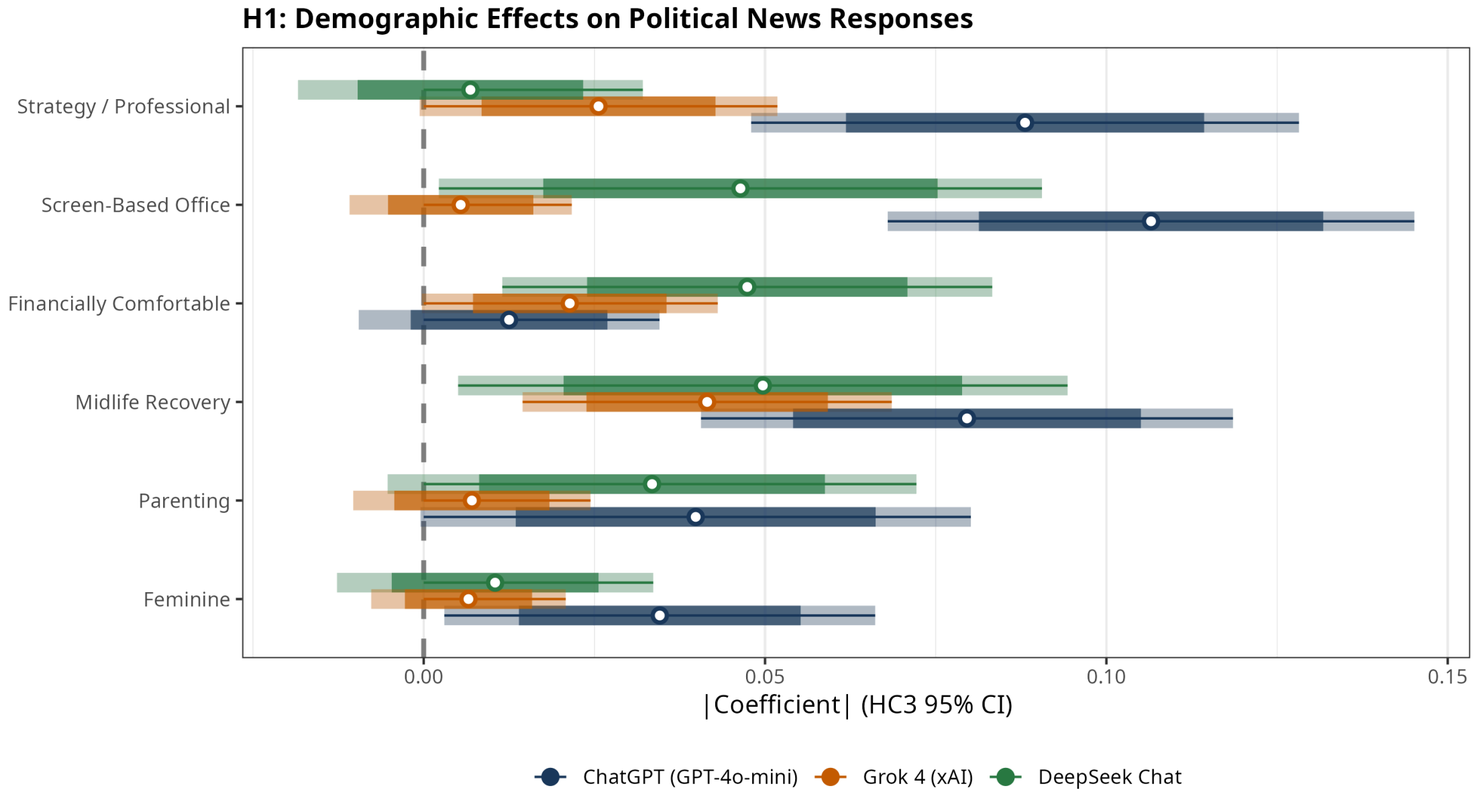
Thanks, that's great. Also, what are the top three news stories currently?

Here are three of the biggest stories right now:

1. Labour leadership challenge.
2. Ebola outbreak declared a global health emergency.
3. Trump pulls back on Iran strike.

Subject: **auditing chatbots**

- Cues measurably shift news round-ups in all three models; ideological cues shift *framing* in Grok and DeepSeek
- Exploratory: tailoring can begin at the search-query stage (GPT-4o-mini, Grok)
- Reliable but small: **passive algorithmic gatekeeping**



The same two roles in the classroom

My research shapes how I teach AI: **advanced and hands-on**, but always critical.

Using AI in advanced ways

AI as measure · across PhD, MSc and undergraduate teaching

- **Applied Language Models (PhD):** fine-tune a classifier and validate it against human coding
- **Computer Programming (MSc):** public APIs, web scraping and LLMs in Python
- **Causal Inference (MSc & PhD):** AI-based measures inside causal designs

Also: Research Design (MSc) · Applied Regression (MSc) · Public APIs with Python (PhD)

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Teaching its limits and implications

AI as subject · ethics and wider politics, built into every course

- **Bias:** an LLM role-plays voters on Brexit; students compare the answers (GV330)
- **Wider costs:** from stereotyping to the energy and politics of data centres
- **Care:** a standing checklist — validity, bias, transparency, privacy

Also: Research Design (MSc) · Applied Regression (MSc) · Public APIs with Python (PhD)

What I would bring to King's

Training students to measure with machines — and to measure the machines.

Needs-led training	Ran the Development Needs Analysis for every ESRC-funded PhD student at LSE
Workshops built for the gaps	Designed workshops for specific needs, open across LSE and other DTPs
Breadth across methods	Text, LLMs, causal inference, research design, programming — UG to PhD
A strong research pipeline	PARTYPRESS, chatbot audits, *Lancet Countdown*: new cases for teaching
Passion for methods	For using *and* teaching computational methods and rigour in social science research

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BACKUP

Additional material

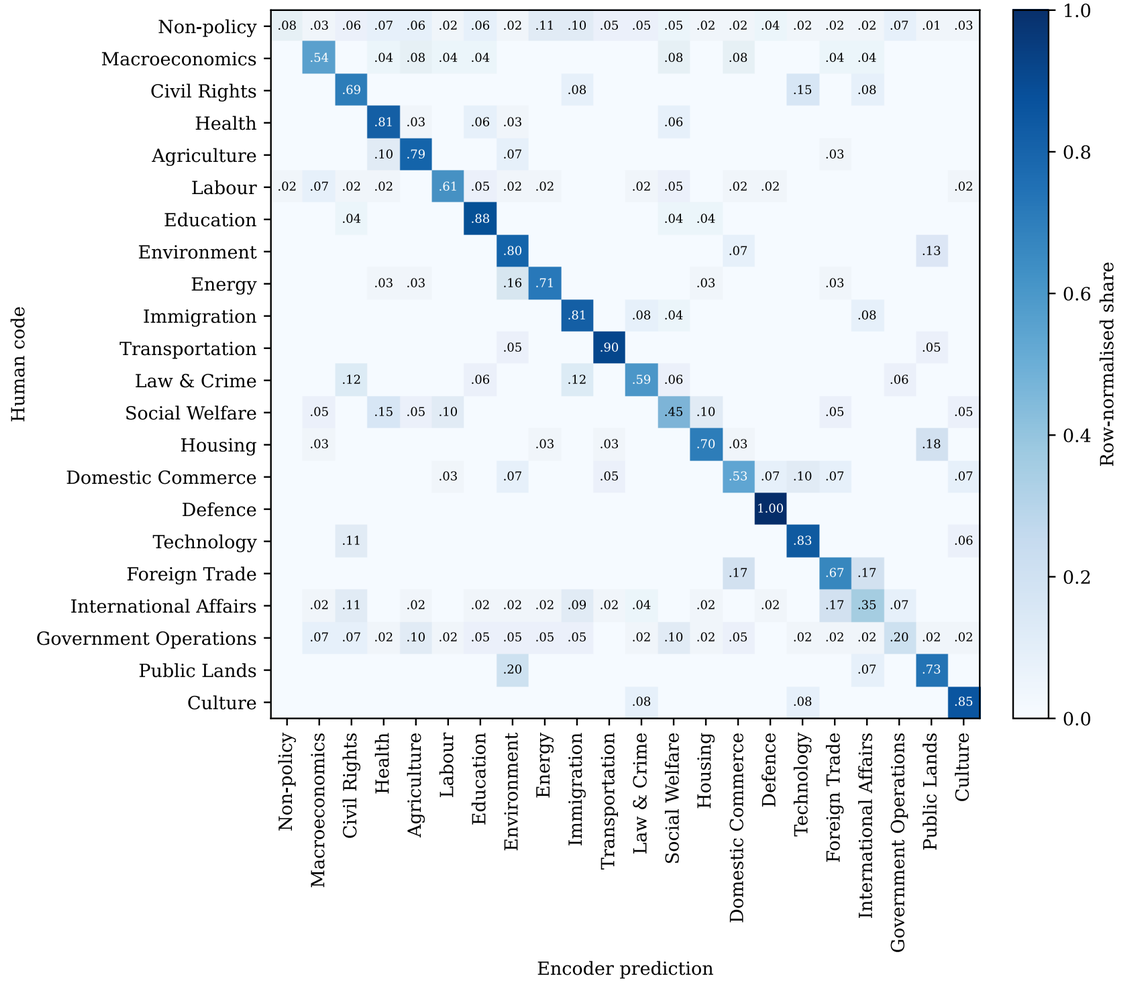
Validating PARTYPRESS against human coders

Design: 2,000 sentences (stratified by predicted code, plus the lowest-confidence); one trained coder; design-weighted. 12 countries not yet covered; multi-coder check under way

Topic: where a sentence is about policy, model and coder agree on the topic of 67% of sentences (κ = 0.65)

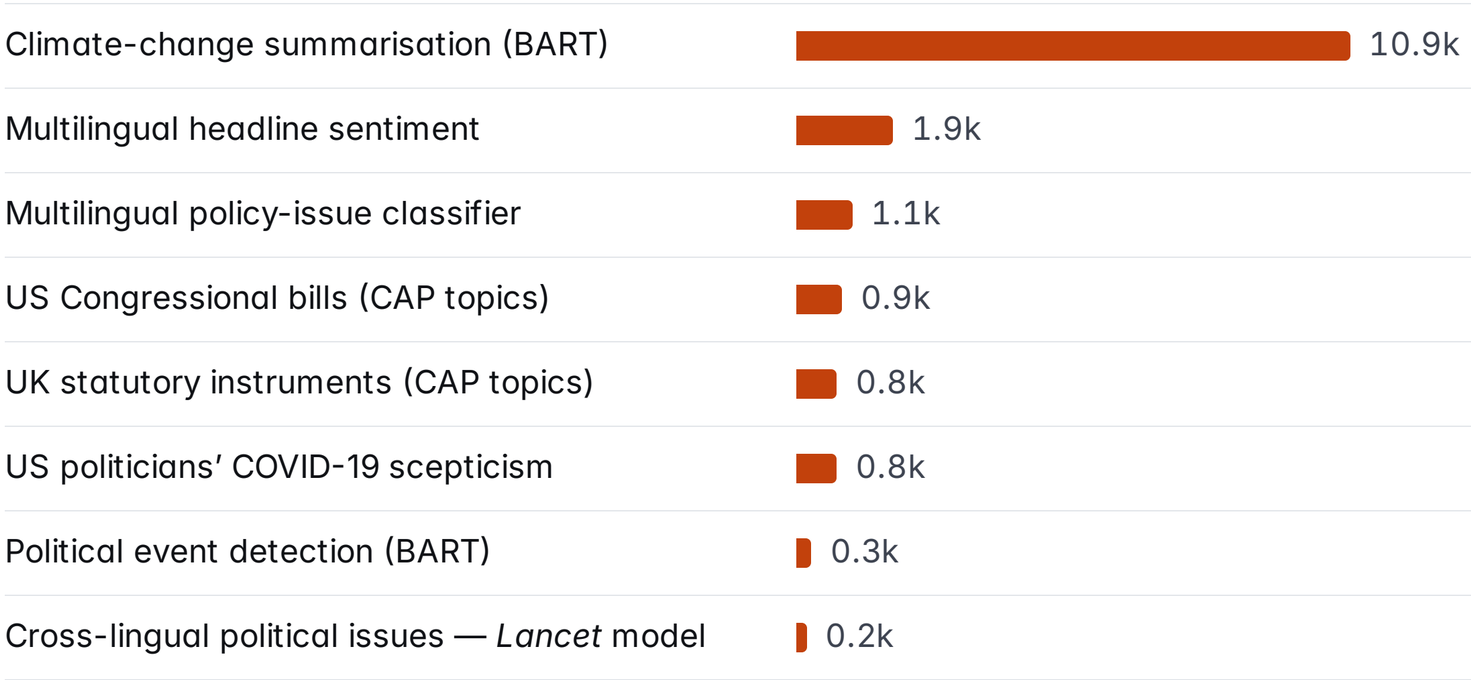
Boundary: the coder places 55% of sentences in the non-policy residual, the model 28% — the main measurement limitation, documented with guidance for users

External: compared with independent issue-attention datasets



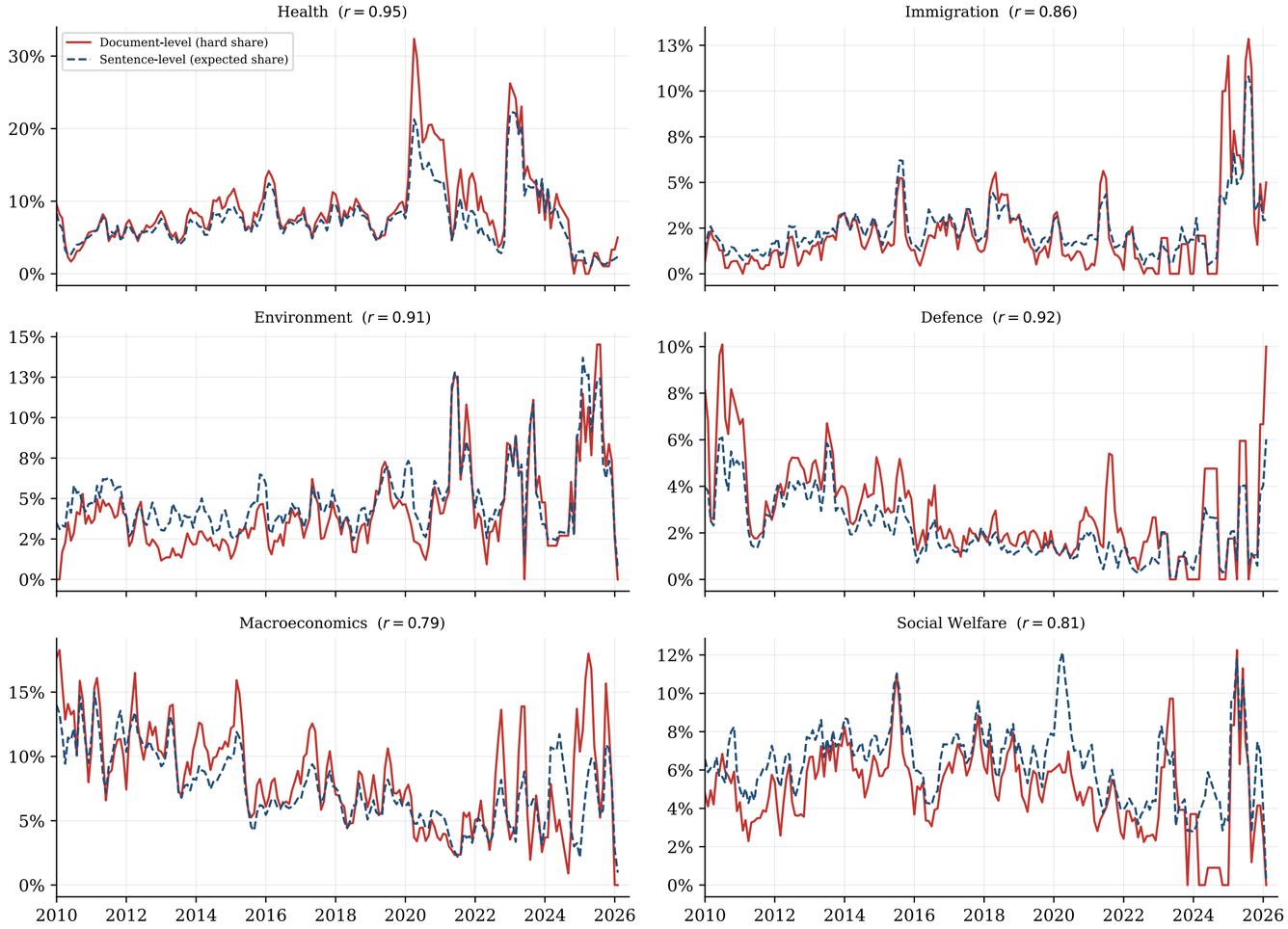
Open models on Hugging Face

Hugging Face MOST DOWNLOADED (ALL TIME)



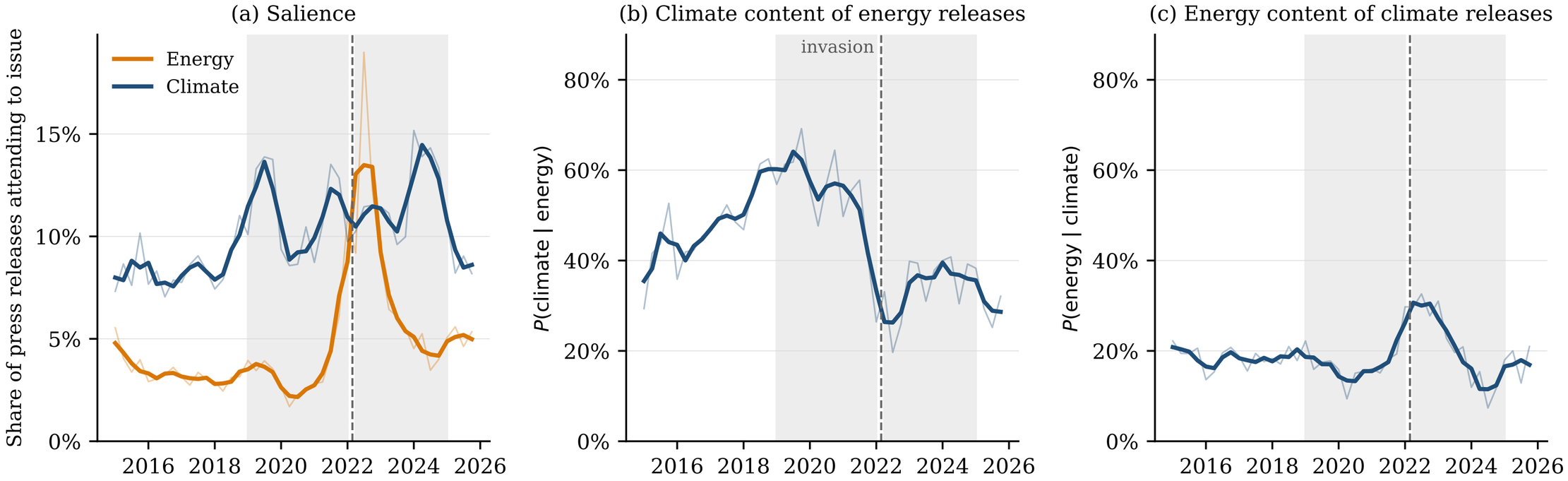
huggingface.co/z-dickson · counts as of October 2026

Sentence-level vs. document-level measurement



Holding the classifier fixed, sentence-level shares yield more reliable party–time series than single document labels

Energy talk grew without climate during the energy crisis



During 2021–22, parties' energy communication grew mainly through releases without climate content

Linking text measures to causal designs

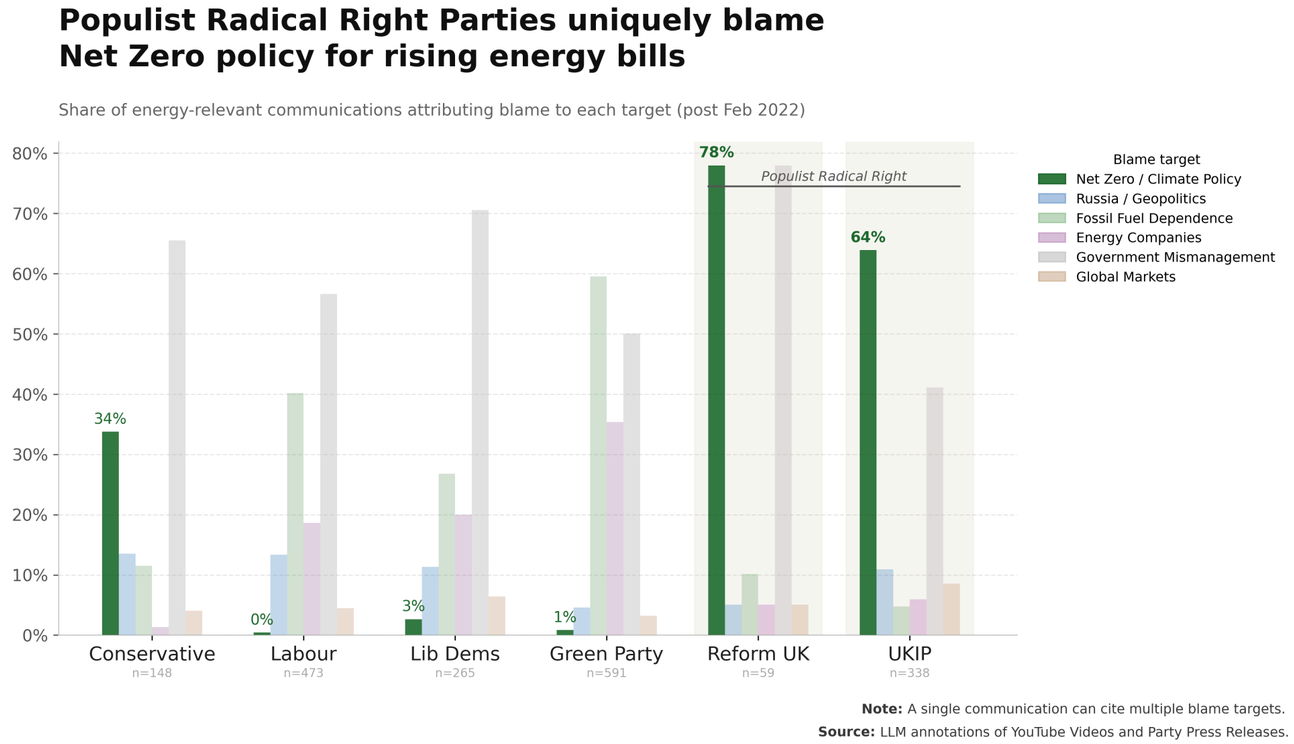
Energy Bills and Manufactured Climate Backlash (British Academy)

Supply: an LLM codes 8,669 press releases & YouTube transcripts by who parties blame for higher bills; validated against a rule-based lexicon

Exposure: 27m energy performance certificates → pre-shock neighbourhood vulnerability

Design: difference-in-differences, 13 waves of Understanding Society

Finding: exposed areas shifted toward the radical right



Note: A single communication can cite multiple blame targets.

Source: LLM annotations of YouTube Videos and Party Press Releases.

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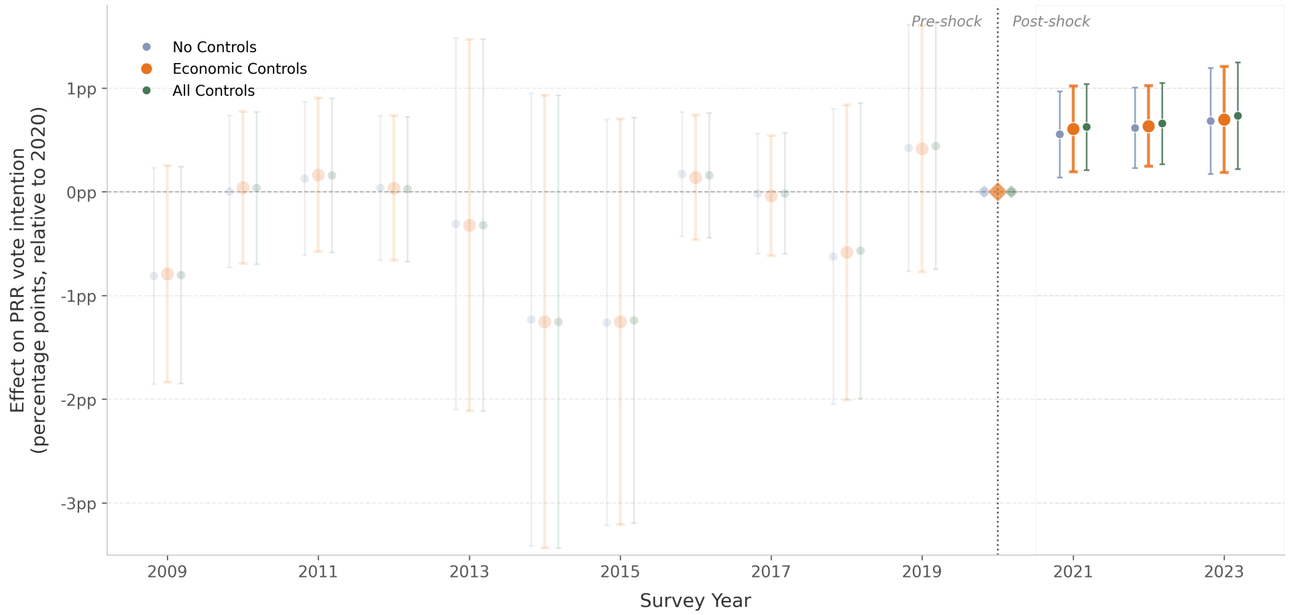
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Energy-inefficient areas increasingly backed Populist Radical Right parties after the 2021 price shock

Effect of high EPC rating on PRR vote intention (percentage points) — three model specifications



Note: reference period 2020. Bars = 95% CI. Dotted line = onset of 2021–22 energy price shock.
Source: Understanding Society (USOC), waves 1–14.

Teaching across levels

Measurement

Applied Language Models (PhD)
Public APIs with Python (PhD)
Computer Programming (MSc)

- Students hand-code, fine-tune a classifier, and validate LLM output against their own codes
- Case studies from my own research — tweets, Congress, party press releases

Design & inference

Causal Inference (MSc & PhD)
Research Design (MSc)
Applied Regression (MSc)

- Potential outcomes, DAGs, difference-in-differences on real data
- Students replicate, reappraise and extend published studies

AI, critically

Data Science Applications to Politics Research (UG, co-taught)

- Generative AI as assistant, measurement tool and data collector — *and* as a subject of study
- A standing checklist: validity, reliability, bias, transparency, privacy